

Dingyang Chen

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Education

- PhD**, Computer Science May 2021 - May 2025
University of South Carolina, Columbia, SC, USA
Advisor: Qi Zhang
- MEng**, Electrical Engineering and Computer Sciences Aug 2018 - May 2019
University of California, Berkeley, Berkeley, CA, USA
- BS**, Computer Science Aug 2014 - May 2018
University of Illinois at Urbana-Champaign, Champaign, IL, USA

Professional Experience

- Amazon**, Seattle, WA Jan 2025 - Present
Applied Scientist
- Fine-tuned large language models to automate product quality detection at Amazon scale.
- Amazon**, Bellevue, WA May 2024 - Aug 2024
Applied Scientist Intern
- Developed multi-armed bandit models for cost-of-goods prediction.
- Siemens**, Remote May 2022 - Aug 2022
Research Intern
- Applied reinforcement learning algorithms to optimize power system reconfiguration.

Publications

Journal Articles

1. **Analyzing Best-Response Dynamics for Cooperation in Markov Potential Games**
Dingyang Chen, Xiaoling Zeng, Thinh T. Doan, Qi Zhang
Transactions on Machine Learning Research (TMLR), 2026.

Conference Proceedings

2. **Natural Policy Gradient for Bayesian Network Policies in Markov Potential Games.**
Dingyang Chen, Zhenyu Zhang, Yuan Ling, Qi Zhang
Reinforcement Learning Conference (RLC), 2026.
3. **Correlated Policy Optimization in Multi-Agent Subteams.**
Dingyang Chen, Jianing Ye, Zhenyu Zhang, Xiaolong Kuang, Xinyang Shen, Ozalp Ozer, Chongjie Zhang, Qi Zhang
International Conference on Learning Representations (ICLR), 2026.
4. **Reinforcement Learning with Euclidean Data Augmentation for State-Based Continuous Control.**
Jinzhu Luo, Dingyang Chen, Qi Zhang
Neural Information Processing Systems (NeurIPS), 2024.

5. **Efficient Sequential Decision Making with Large Language Models.**
Dingyang Chen, Qi Zhang, Yinglun Zhu
 Empirical Methods in Natural Language Processing (EMNLP), 2024.
6. **E(3)-Equivariant Actor-Critic Methods for Cooperative Multi-Agent Reinforcement Learning.**
Dingyang Chen, Qi Zhang
 International Conference on Machine Learning (ICML), 2024.
7. **Context-Aware Bayesian Network Actor-Critic Methods for Cooperative Multi-Agent Reinforcement Learning.**
Dingyang Chen, Qi Zhang
 International Conference on Machine Learning (ICML), 2023.
8. **Communication-Efficient Actor-Critic Methods for Homogeneous Markov Games.**
Dingyang Chen, Qi Zhang
 International Conference on Learning Representations (ICLR), 2022.

Refereed Workshop Papers

9. **From Business Metrics to Behavioral Personas Controllable User Simulation for Pre-Deployment Agent Testing.**
 Zhenyu Zhang, Yuan Ling, Dingyang Chen, Xinyang Shen
 Workshop on Agents in the Wild at ICML, 2026.
10. **Tree of Options: Temporally Extended World Modeling, Planning, and Execution with Large Language Models.**
 Xiaoling Zeng, Dingyang Chen, Qi Zhang
 Workshop on World Models at ICLR, 2026.
11. **Convergence Rates of Bayesian Network Policy Gradient for Cooperative Multi-Agent Reinforcement Learning.**
Dingyang Chen, Zhenyu Zhang, Xiaolong Kuang, Xinyang Shen, Ozalp Ozer, Qi Zhang
 Workshop on Bayesian Decision-making and Uncertainty at NeurIPS, 2024.
12. **Convergence and Price of Anarchy Guarantees of the Softmax Policy Gradient in Markov Potential Games.**
Dingyang Chen, Qi Zhang, Thinh T. Doan
 Decision Awareness in Reinforcement Learning Workshop at ICML, 2022.
13. **A Meta-Gradient Approach to Learning Cooperative Multi-Agent Communication Topology.**
 Qi Zhang, Dingyang Chen
 Workshop on Meta-Learning at NeurIPS, 2021.

Professional Service

Conference Reviewer

ICML 2024 & 2025 & 2026, AISTATS 2026, AAAI 2025, ICLR 2024, NeurIPS 2023, 2025, and the NeurIPS 2022 Deep RL Workshop

Journal Reviewer

IEEE TNNLS

Teaching Experience

Graduate Teaching Assistant - Visualization Tools, University of South Carolina [FA24]
Graduate Teaching Assistant - Algorithmic Design I, University of South Carolina [FA23]